

TriageAI-X

An Explainable Multimodal Clinical Decision Support System for
Emergency Department Triage



Project Proposal

Submitted for the Triagegeist Hackathon

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1. Introduction

Emergency department (ED) triage is one of the most critical processes in modern health-care systems. Triage clinicians must rapidly assess patient severity and prioritize treatment under extreme cognitive load, incomplete information, overcrowded conditions, and limited staffing.

Traditional triage systems such as the Emergency Severity Index (ESI) and Manchester Triage System (MTS) rely heavily on human judgment. However, studies have demonstrated substantial inter-rater variability and risks of undertriage, particularly among vulnerable patient populations.

Artificial Intelligence (AI) offers the potential to improve consistency, identify overlooked high-risk cases, and support clinicians through interpretable decision support systems.

This project proposes the development of **TriageAI-X**, an explainable multimodal emergency triage intelligence system that combines structured clinical features, patient history, and free-text chief complaints to predict emergency acuity levels and support clinical decision-making.

2. Problem Statement

Current emergency triage systems face several limitations:

- High variability in clinician decision-making
- Risk of delayed treatment due to undertriage
- Increasing emergency department overcrowding
- Limited integration of historical and textual patient data
- Lack of explainability in automated clinical systems

The goal of this project is to develop an AI-powered triage assistance framework capable of:

- Predicting patient triage acuity levels
- Understanding chief complaint narratives using NLP
- Identifying high-risk patients
- Providing interpretable clinical explanations
- Detecting possible demographic bias in triage decisions

3. Objectives

The primary objectives of the proposed system are:

1. Develop a multimodal AI model for emergency triage prediction.

2. Integrate structured physiological data and free-text complaints.
3. Improve identification of critically ill patients.
4. Provide explainable AI outputs using SHAP.
5. Analyze fairness and potential triage bias across patient groups.
6. Build an interactive clinical decision support prototype.
7. Create a fully reproducible Kaggle notebook and research writeup.

4. Dataset Description

The project utilizes the synthetic emergency department triage dataset released for the Triagegeist competition.

4.1. Dataset Files

File	Description
train.csv	Training dataset containing features and triage acuity labels
test.csv	Test dataset without target labels
chief_complaints.csv	Free-text chief complaint narratives
patient_history.csv	Binary comorbidity and medical history features
sample_submission.csv	Submission format example

Table 1: Dataset Files

4.2. Target Variable

The target variable is:

$$\text{triage_acuity} \in \{1, 2, 3, 4, 5\}$$

where:

- 1 = Most urgent
- 5 = Least urgent

5. Methodology

The proposed methodology consists of multiple stages.

5.1. Stage 1: Data Integration

The following datasets will be merged using `patient_id`:

- `train.csv`
- `chief_complaints.csv`
- `patient_history.csv`

5.2. Stage 2: Data Preprocessing

Preprocessing steps include:

- Missing value handling
- Outlier detection
- Encoding categorical variables
- Feature normalization
- Text cleaning and tokenization

5.3. Stage 3: Feature Engineering

Clinical feature engineering will include:

- Shock Index
- Pulse Pressure
- NEWS-like severity indicators
- Tachycardia indicators
- Hypoxia flags
- Comorbidity burden score

Example:

$$\text{Shock Index} = \frac{\text{Heart Rate}}{\text{Systolic Blood Pressure}}$$

5.4. Stage 4: Natural Language Processing

Free-text chief complaints will be processed using NLP techniques.

Approaches include:

- TF-IDF Vectorization
- ClinicalBERT embeddings
- Risk phrase extraction
- Complaint clustering

5.5. Stage 5: Machine Learning Models

The following models will be evaluated:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM
- CatBoost

An ensemble approach will be developed for improved performance.

6. Explainable AI

Healthcare AI systems require transparency and interpretability.

The project will implement SHAP (SHapley Additive exPlanations) to:

- Explain model predictions
- Identify important clinical features
- Improve clinician trust
- Visualize patient-specific risk factors

7. Fairness and Bias Analysis

The project will evaluate whether systematic triage bias exists across:

- Age groups
- Sex
- Insurance categories
- Language groups

Bias evaluation metrics include:

- Group-wise recall
- False negative rate
- Calibration analysis
- Fairness disparity measures

8. Evaluation Metrics

Model performance will be evaluated using:

Metric	Purpose
Accuracy	Overall prediction correctness
Macro F1 Score	Balanced class evaluation
Quadratic Weighted Kappa	Ordinal classification evaluation
Recall for ESI-1	Critical patient identification
Confusion Matrix	Error analysis

Table 2: Evaluation Metrics

9. System Architecture

The proposed system architecture includes:

1. Data ingestion
2. Data preprocessing
3. Feature engineering
4. NLP processing
5. Ensemble modeling
6. Explainability layer
7. Fairness analysis
8. Dashboard deployment

10. Deployment Plan

A Streamlit-based interactive dashboard will be developed.

Features include:

- Patient intake interface
- Acuity prediction visualization
- SHAP explanation dashboard
- Bias analysis reports
- Risk monitoring interface

11. Expected Outcomes

The proposed system is expected to:

- Improve triage consistency
- Identify high-risk patients earlier
- Reduce undertriage risk

- Provide clinically interpretable predictions
- Demonstrate fairness-aware AI methodology

12. Future Scope

Future enhancements may include:

- Real-world hospital validation
- Deep learning transformer models
- Real-time monitoring systems
- Integration with electronic health records
- ICU admission prediction

13. Conclusion

This project proposes a clinically relevant and technically rigorous AI-powered emergency triage intelligence framework. By integrating structured physiological data, patient history, and natural language processing, the proposed system aims to improve emergency department triage accuracy, interpretability, and fairness.

The final system will provide an explainable and reproducible proof-of-concept capable of informing future clinical decision support research and deployment.

14. References

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